

Interim Project Update

Insurance Sales RL Agent

Emotion-aware strategy selection with PPO

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Problem Statement and Objectives

Which sales strategy should the agent choose next?

Buyer intent changes across turns.
Emotion and objections affect the response.
Existing cover affects product suitability.

Research question

Can PPO improve strategy selection while maintaining suitability and low-pressure behavior?

Adaptive action selection

Use structured customer state rather than a fixed conversation sequence.

Controlled product information

Filter product facts before the response generator receives them.

Reproducible evaluation

Compare PPO with random, rule and supervised policies using one simulator.

Literature Survey

Paper (short title)	Verified contribution	Connection to this project
Concurrent bilateral RL Bagga et al. 2020 [1]	Actor-critic RL with supervised pre-training for negotiation.	Supervised initialization before PPO training.
Adaptive RL switching Sengupta et al. 2021 [2]	Opponent classification and adaptive strategy switching.	Profile, emotion and objection features guide actions.
General negotiation RL Renting et al. 2024 [3]	Graph-based RL policy for transfer across negotiation problems.	Highlights the limits of our fixed-size linear policy.
Persuasion Games Karande et al. 2024 [4]	LLM persuasion with auxiliary retrieval and fact validation.	Supports separate strategy selection and response wording.
AgenticPay Liu et al. 2026 [5]	Benchmark for language-based buyer-seller negotiation.	Motivates a direct-LLM baseline and multi-turn evaluation.

Project focus: constrained sales dialogue actions. Claim-settlement examples remain analogies.

Implemented Approach

Component	Responsibility	Current implementation
PPO policy	Choose the next dialogue action.	Masked linear actor-critic in NumPy. 16 controlled actions.
Product / rule engine	Control eligible product facts.	Need, age and budget filters. 8 synthetic product records.
Response generator	Express the assigned strategy.	Local Ollama Llama 3.2 3B. Verified templates as fallback.

Live action constraints

Clarify ambiguous amounts and payment questions.
Record any change to the policy proposal.

Response checks

Check numeric claims, role and repetition.
Exact premiums require a verified insurer quote.

Dataset and Customer Understanding

1,200

Synthetic conversations

20,396

Dialogue turns

3,000

Synthetic profiles

Split	Train	Validation	Test
Conversations	812	164	224

NLP and entities

Naive Bayes: intent, emotion, objection, stage.
Rules: needs, budget, coverage and stop cues.

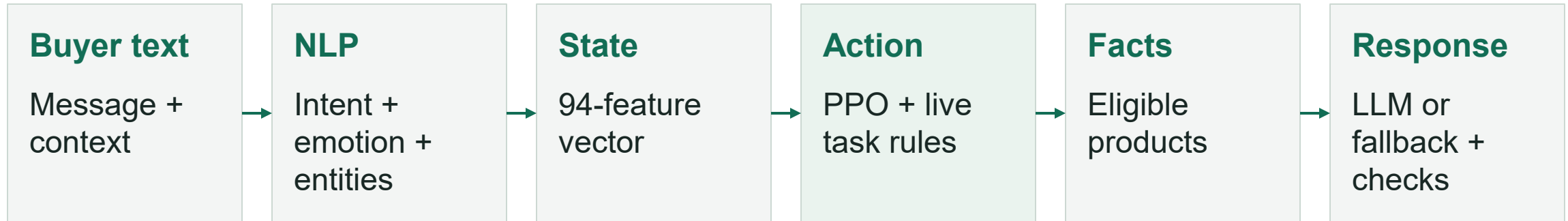
Current evidence gap

Emotion accuracy: 53.7%. Stage: 37.0%.
Both measure a held-out synthetic test set.

Split by scenario and transcript group to reduce leakage. Human-customer generalization remains untested.

System Pipeline

Live conversation



Conversation memory carries the response and buyer context into the next turn.

Offline learning



Updated policy parameters drive the next simulated rollout. Live chat uses the saved checkpoint.

RL Formulation

State: 94 features

Intent, emotion, need, objection and stage.
Profile, budget, coverage and recent actions.
Normalized trust, engagement and purchase intent.

Action space: 16 strategies

Discovery: needs, clarification, coverage gap.

Education: product, value, risk, affordability, personalization and comparison.

Concerns: objection handling, yes-yes, trust.

Outcomes: alternatives, commitment, follow-up and respectful rejection.

Selected reward event	Weight
Suitable purchase	+5.0
Qualified lead	+1.0
Suitable product	+0.5
Pressure	-1.0
Unsupported claim	-2.0
Unsuitable recommendation	-3.0

Action masks enforce eligibility and stop requests.
Maximum episode length: 20 turns.

PPO Implementation

$$L_{\text{clip}} = \text{mean}[\min(r A, \text{clip}(r, 0.8, 1.2) A)]$$

r = current-policy probability / rollout-policy probability. A = estimated advantage.

Portable NumPy actor-critic

Rollout Masked sampling with saved log probabilities.

Advantage GAE with raw critic targets and normalized actor advantages.

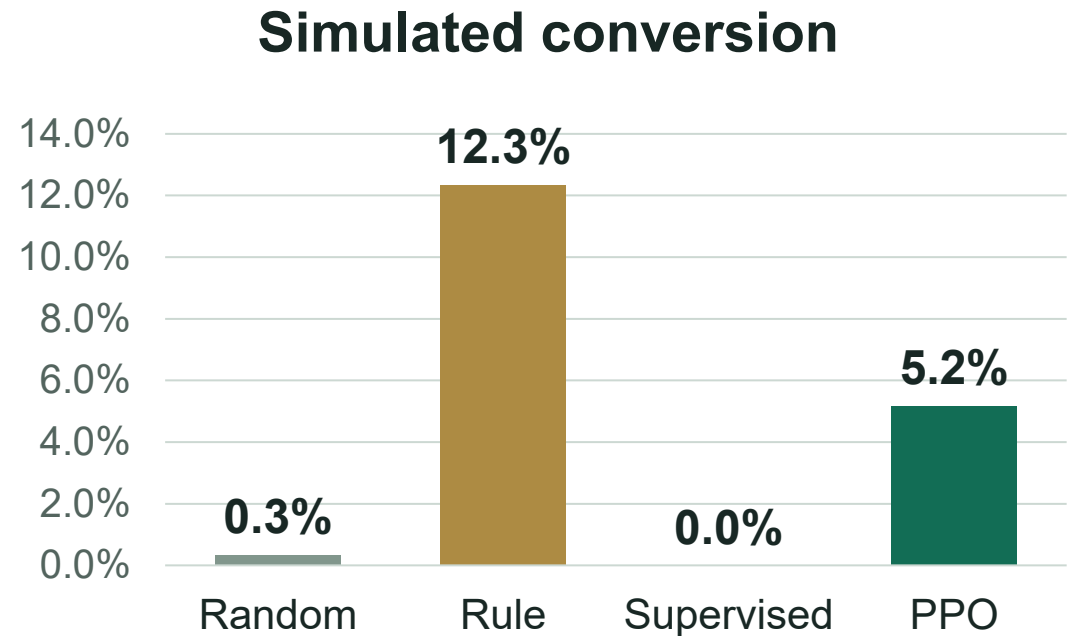
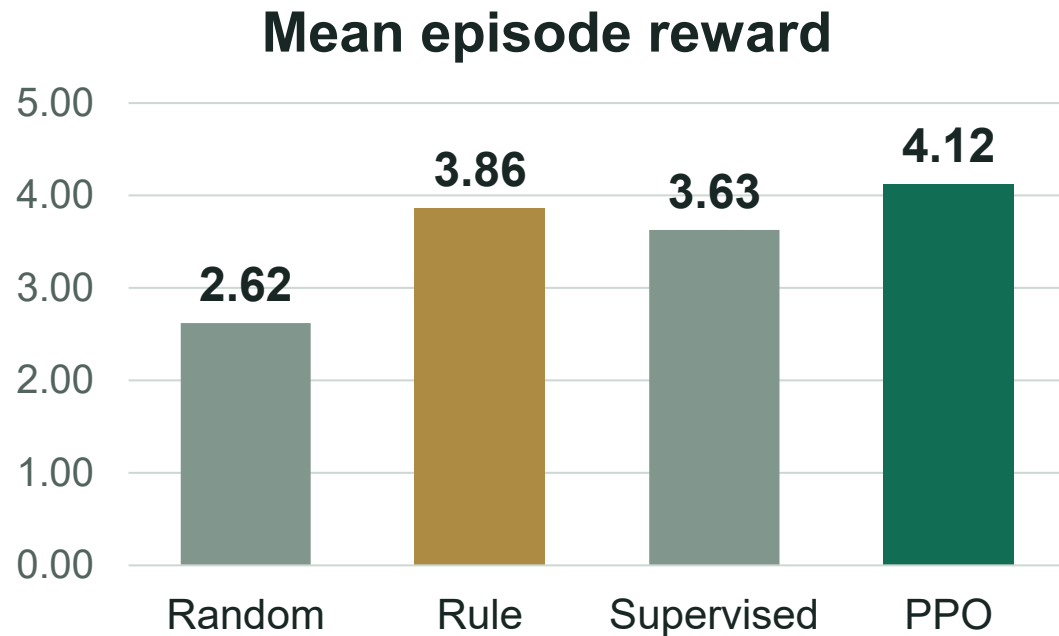
Update Clipped objective, entropy bonus, Adam and gradient clipping.



20,000 steps per seed. Rollout 1,024. Batch 256. Four epochs. Learning rate 0.0003.

Evaluation Results

Three seeds. 200 simulated evaluation episodes per seed. Point estimates below.



PPO has higher reward. The rule policy converts more often.

Objection resolution: PPO 78.2%, rule 69.7%. Six retrained ablations test feature and reward choices.

Synthetic simulation only. Wide intervals across three seeds do not establish statistical superiority.

Demonstration and Remaining Work

The screenshot displays the 'Insurance Sales Agent' interface, a research demo. The top navigation bar includes 'Conversation', 'Catalogue', and 'Results' tabs, along with a 'New session' button. The main interface is divided into several sections:

- Buyer profile (SIMULATED):** Includes a 'Family-oriented buyer' (Age 35 / Low budget) and a 'Decision policy' set to 'PPO'.
- Conversation (LIVE SESSION):** Shows a chat window with a message from the 'Insurance agent' (01:04 am, Local LLM) stating: "I don't have the ability to provide a specific payment amount as our synthetic catalogue only offers premium bands. A verified insurer quote would be needed to determine the exact premium for your family's health insurance needs." The chat also shows a 'what will be my payment' input field and a 'Basic checks passed' status.
- Decision trace (TURN 03):** Displays the 'SELECTED ACTION' as 'Affordability Framing' and the 'Understanding' section with 'Intent: Price Inquiry', 'Emotion: Concerned', and 'Objection: None'.
- Buyer state:** Lists attributes such as 'Need: Family Health', 'Budget: Low', 'Existing cover: Unknown', 'Stage: Evaluation', and 'Buyer amount: 250000 (Unknown)'.
- Response checks:** Shows 'Basic response checks passed' and 'Source: Local LLM'.

Local text demo

Conversation, product catalogue and decision trace.
<http://127.0.0.1:8765/>

Current milestone

Trained PPO, saved evaluation and local LLM conversation.

Next research steps

Stronger emotion/stage models.
Direct-LLM and human evaluation.
Voice input/output later.

Research scope

Synthetic products only.
No policy or payment flow.